01

Child Malnutrition

Risk in Developing Countries

Which children are most at risk — and what does the data tell us?

📊 UNICEF JME 2025 Database | 154 Countries | Official UN Data

154
Countries

5
Indicators

25
Crisis Zone

PROBLEM STATEMENT

Millions of children in developing countries are stunted, wasted, or overweight. Many governments do not know which regions need help most — or when the problem is getting worse.

WHY IT MATTERS

🌍 45 million children suffer from wasting globally

📉 Stunted growth causes lifelong damage to health

🏥 Malnutrition is linked to half of all child deaths

🎯 South Asia and West Africa are the hardest hit

🔗 data.unicef.org | UNICEF JME 2025 Dataset

02

Objectives

Five goals that guided our analysis of child malnutrition across 154 countries

01

Group Countries by Malnutrition Type

Use K-Means clustering on 5 UNICEF indicators — stunting, wasting, overweight, and co-occurrence — to find 4 natural country groups

02

Identify the Crisis Zone Countries

Find which 25 countries have the worst malnutrition profiles — highest stunting and wasting — so aid can be directed where it is needed most

03

Understand What Drives Risk

Use Random Forest to find which factors — region, year, survey type — best predict whether a country is at high malnutrition risk

04

Predict High Risk Countries

Train a model on real UNICEF survey data (80/20 split) to correctly flag countries at high risk before conditions worsen further

05

Turn Findings into Simple Recommendations

Translate cluster profiles and risk predictions into clear actions that governments and aid organisations can act on immediately

03

Approach & Methodology

UNICEF Data
154 countries
5 indicators

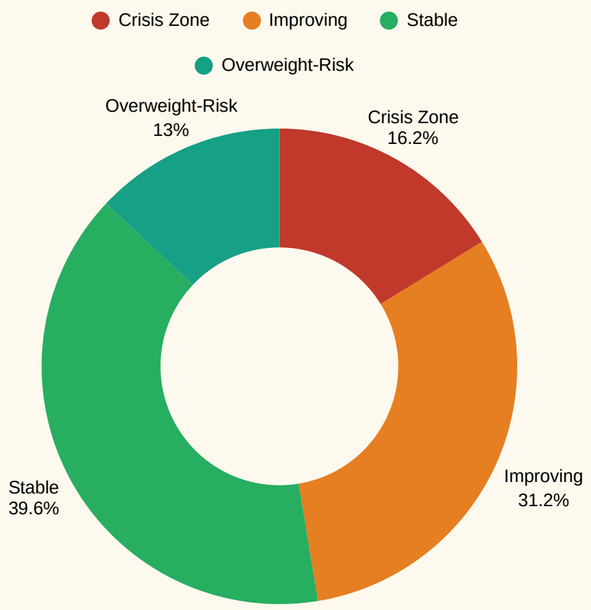
Clean & Scale
Mean impute
StandardScaler

K-Means K=4
Elbow method
Clusters

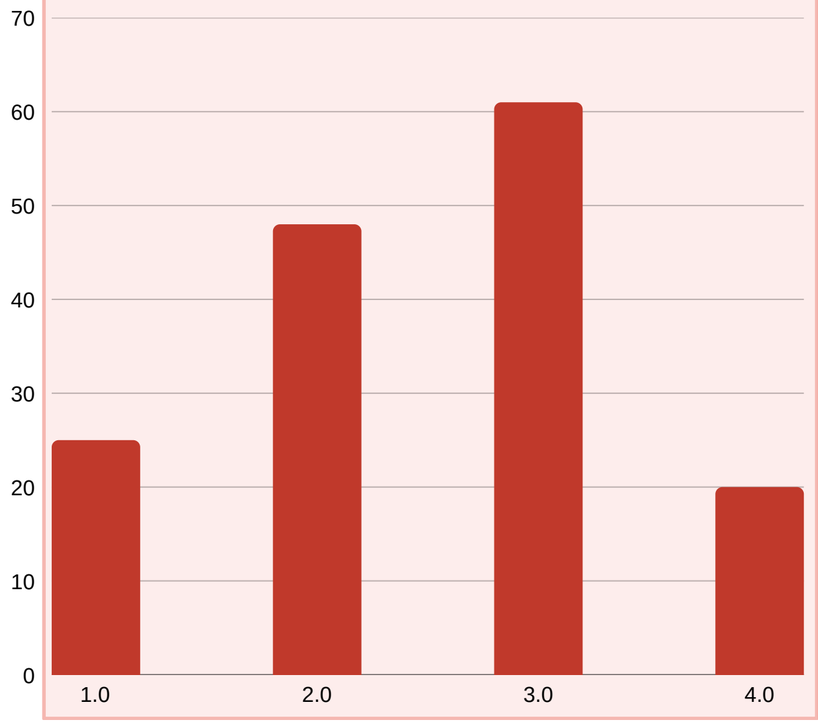
Random Forest
80/20 split
Predict risk

Risk Insights
Top drivers
High-risk list

Countries by Risk Level



Cluster Sizes (No. of Countries)



Cluster Profiles (K=4)

Crisis Zone
25 countries
High stunting & wasting
South Asia + West Africa
Needs urgent action

Improving
48 countries
Moderate malnutrition
Getting slowly better
Needs ongoing support

Stable
61 countries
Low malnutrition overall
Better health systems
Maintain current progress

Overweight-Risk
20 countries
Low wasting, high overweight
Nutrition transition phase
Different challenge

Top Risk Predictors: Survey Year (48.5%) | South Asia Region (25.2%) | West & Central Africa (9.8%)

04 Impact & Social Importance

Economic Advantages

Stops Wasting Aid Money

Without this analysis money is spread equally across all 154 countries. Now we know the exact 25 that need it most — so every dollar goes where it saves the most lives.

Prevention is Cheaper Than Treatment

Treating a severely malnourished child in hospital costs far more than stopping it early. The model gives an early warning so help arrives before the crisis becomes expensive.

A Healthy Child Earns More as an Adult

Stunted children grow up to earn less, miss more work, and need more healthcare. Solving malnutrition early is an investment that pays back for 50 years.

Reduces Long Term Dependency on Aid

Countries that solve malnutrition become more self-sufficient over time. They need less international support — freeing up global funds for other crises elsewhere.

Social Impact

Children Live Longer, Healthier Lives

When malnutrition is caught early and treated, children grow up stronger. They do better in school and live healthier adult lives.

Mothers and Families Are Protected

Malnourished mothers have malnourished babies. Helping one person in a family can protect the whole next generation.

Children Can Go to School

A hungry or sick child cannot learn. Reducing malnutrition means more children can attend school, finish their education, and escape poverty.

Poorer Countries Can Grow Stronger

Malnutrition holds entire countries back. A healthy population works more, earns more, and helps the whole nation develop faster.

Supports the UN's Global Goals

This work directly supports UN SDG Goal 2 — Zero Hunger. Every child saved from malnutrition is a step closer to a fairer world for everyone.

Key Insight: The 25 Crisis Zone countries — mostly in South Asia and West Africa — need the most urgent help. Getting aid there first saves the most lives.